

EE2 Circuits and Systems

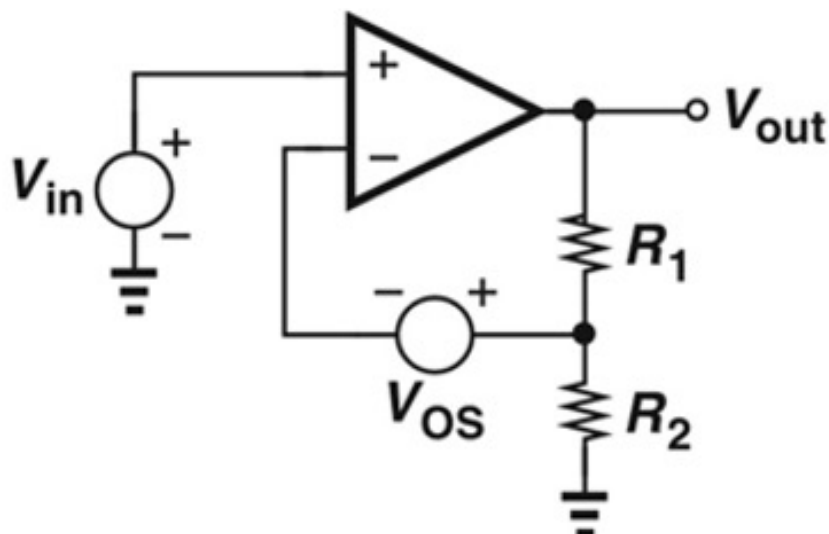
Problem Class 2 (Sheet 1 con't)

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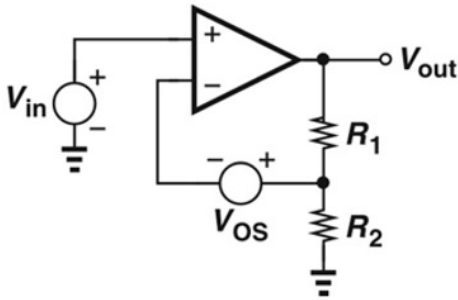
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Test Yourself - Problem 1 (Sheet 1 Q4)

In the non-inverting amplifier shown below, the op-amp offset at its input is represented by a voltage source in series with the inverting input. Calculate V_{out} .



Test Yourself - Solution 1 (Sheet 1 Q4)



- Real op-amps have another imperfection known as input offset voltage V_{OS} .
- MCP601 has a worst case V_{OS} of $\pm 3\text{mV}$
- Principle of superposition
- Consider V_{in} and V_{OS} separately:

Assume $V_{OS} = 0$,
$$V_{out1} = V_{in} \left(1 + \frac{R_1}{R_2}\right)$$

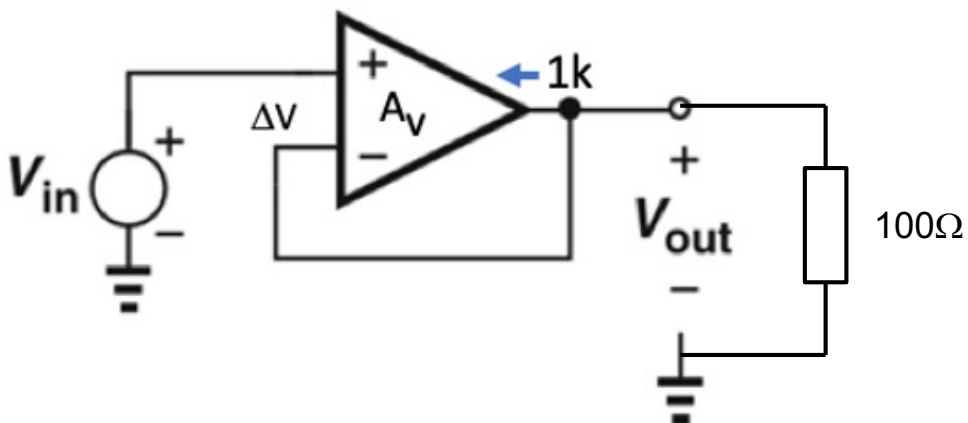
Assume $V_{in} = 0$, voltage across R_2 is V_{OS} .

$$V_{OS} = V_{out2} \left(\frac{R_2}{R_1 + R_2}\right)$$

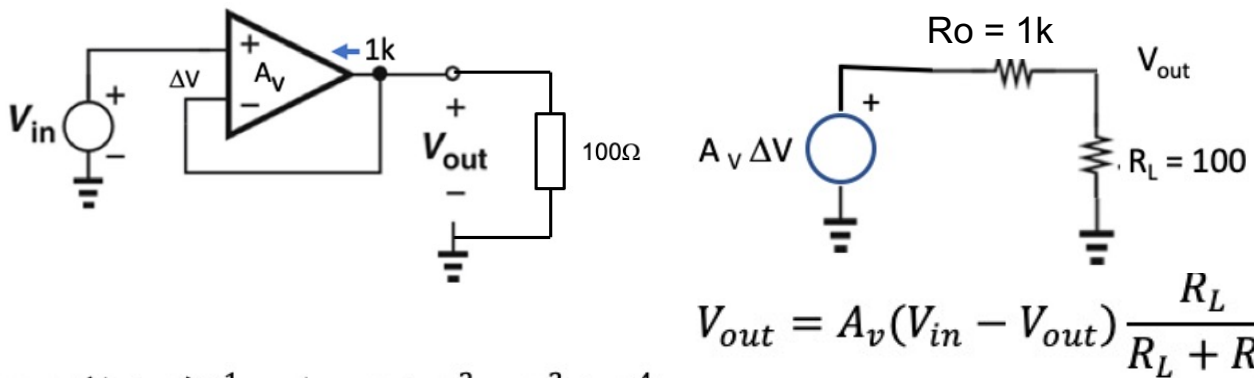
$$V_{out} = (V_{in} + V_{OS}) \left(1 + \frac{R_1}{R_2}\right)$$

Explain it - Problem 2 (Sheet 1 Q5)

The unity-gain buffer shown below is designed to drive a 100Ω load with an acceptable gain error of 0.5%. Determine the required op-amp open-loop gain A_v if the op-amp has an output impedance of $1\text{k}\Omega$.



Explain it - Solution 2 (Sheet 1 Q5)



Since $(1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$

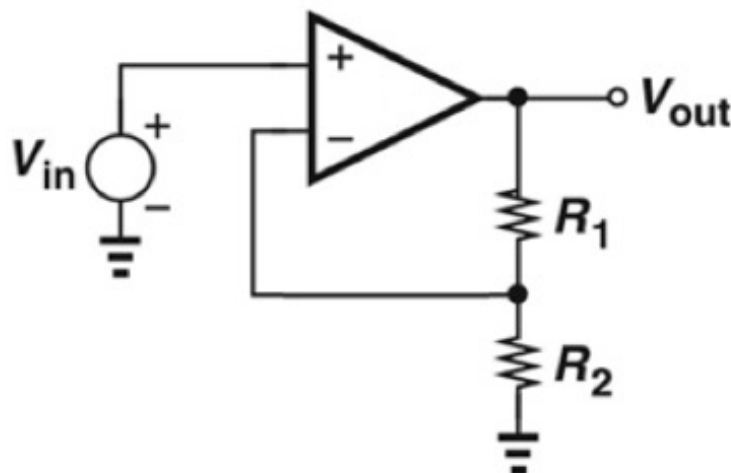
$$\frac{V_{out}}{V_{in}} = \frac{1}{1 + \frac{R_L + R_o}{A_v R_L}} \approx 1 - \frac{R_o + R_L}{A_v R_L} = 1 - \text{error (for } A_v \gg 1)$$

$$\left| \frac{R_o + R_L}{A_v R_L} \right| = 0.005 \quad A_v = \left| \frac{R_o + R_L}{0.005 R_L} \right| = \frac{1000 + 100}{0.005 * 100} = 2200$$

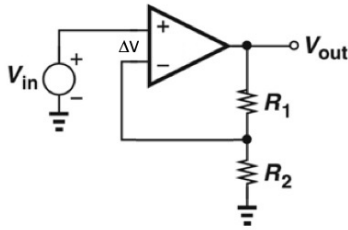
Test Yourself - Problem 3 (Sheet 1 Q6)

The non-inverting amplifier has a gain of 4 within 0.2% accuracy. A_v is finite, otherwise op-amp is ideal.

Given that $R_1 + R_2 = 20\text{k}\Omega$, determine the minimum value of A_v .



Test Yourself - Solution 3 (Sheet 1 Q6)



Ideal gain $(1 + R_1/R_2) = 4$, and $R_1 + R_2 = 20k$

Hence, $R_1 = 15k$ and $R_2 = 5k$.

Unfortunately the gain is not exactly 4 due to the finite A_v .

Instead, the gain is:

$$\frac{V_{out}}{V_{in}} = \frac{A_v}{1 + A_v \frac{R_2}{R_2 + R_1}} = \left(1 + \frac{R_1}{R_2}\right) \left(\frac{1}{1 + \left(1 + \frac{R_1}{R_2}\right) \times \frac{1}{A_v}} \right)$$

$$\left[\text{Since } (1 + x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots \right] \approx \left(1 + \frac{R_1}{R_2}\right) \left[1 - \left(1 + \frac{R_1}{R_2}\right) \frac{1}{A_v} \right]$$

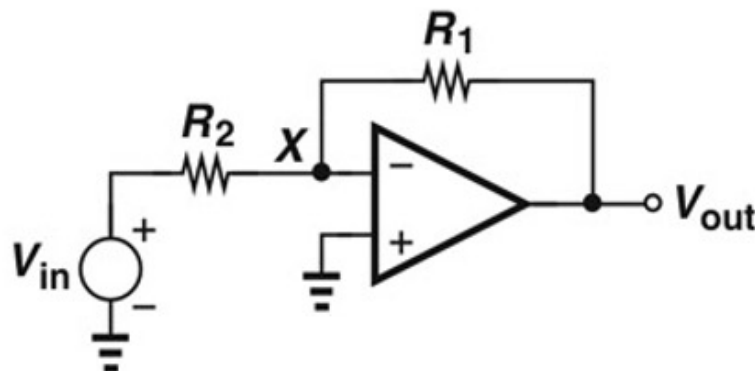
Since we know $\left(1 + \frac{R_1}{R_2}\right) = 4$, the nominal gain, and the gain error is 0.2%,

$$\left(1 + \frac{R_1}{R_2}\right) \frac{1}{A_v} = 0.002$$

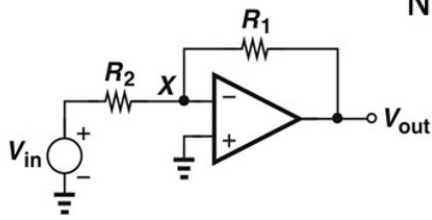
$$\text{Hence, } A_v = 4/0.002 = 2000.$$

Test yourself - Problem 4 (Sheet 1 Q7)

An inverting amplifier shown below with a gain of -8 with a gain error of $\pm 11\%$ is required. What are the tolerance of the resistors R_1 and R_2 , and what is the minimum open-loop gain of the op-amp required? You may assume that R_{out} and the input current of the op-amp are negligible. State any other assumption used.



Test yourself - Solution 4 (Sheet 1 Q7)



Nominal gain = $R_1/R_2 = -8$, Gain error = $\pm 11\%$.

R_1 and R_2 needs to be at least accurate to $\pm 5\%$.

Check: $R_1 \rightarrow +5\%$, $R_2 \rightarrow -5\%$, Absolute gain error = $+10.5\%$

$R_1 \rightarrow -5\%$, $R_2 \rightarrow 5\%$, -9.5% .

Assume that A_v is high enough for the equation $G = -R_2/R_1$ applies. Now assume 0.5% is available as gain error due to A_v being finite. What is the minimum value of A_v ?

$$V_x = -V_{out}/A_v \quad \frac{V_{in} - V_x}{R_2} = \frac{V_x - V_{out}}{R_1}$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_1}{R_2} \frac{1}{1 + \frac{1}{A_v} \left(1 + \frac{R_1}{R_2}\right)} \simeq -\frac{R_1}{R_2} \left[1 - \frac{1}{A_v} \left(1 + \frac{R_1}{R_2}\right)\right]$$

$$|error| < 0.5\% = \left| \frac{1}{A_v} \left(1 + \frac{R_1}{R_2}\right) \right| \simeq \frac{1}{A_v} (1 + 8)$$

$A_v > 1800.$